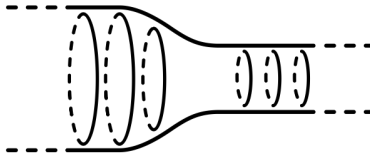


MAE 3127 - Fluid Mechanics Laboratory

Laboratory-1

(Total points = 100)



Created by Oleksandr Panasovskyi
from Noun Project

Figure 1: flow pressure by Oleksandr Panasovskyi from flow pressure Icons, Noun Project (CC BY 3.0)

COMPLETE THIS PORTION BEFORE STARTING YOUR EXPERIMENT:

You will be working in small groups during this experiment. While the data gathering is done collectively, your analysis will be done individually. Put down all the names of your group members and have each of them sign off on this portion, acknowledging the group work. Without their signatures, this portion will not be graded.

Last Name	First Name	Student's Signature	Date.
_____	_____	_____	_____
_____	_____	_____	_____
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_____	_____	_____	_____

Laboratory-1: Venturi Tube

Nomenclature

d	diameter (mm)
g	gravitational acceleration (9.81 m s^{-2})
h	measured flow head (mm)
l	distance along Venturi tube (mm)
u	flow velocity (m s^{-1})
Q	volumetric flow rate ($\text{m}^3 \text{ s}^{-1}$)

Subscripts

1	inlet related property
2	throat related property
n	n^{th} location related property

Objectives

1. Gain fundamental understanding of the Bernoulli equation
2. Perform experiments on a Venturi contracting-expanding tube
3. Understand possible sources for error and differences between theory and measurements
4. Gain basic experience with a manometer apparatus

Theory

The basic underlying Bernoulli equation states that

$$p + \frac{1}{2}\rho u^2 + \rho gh = \text{constant} . \quad (1)$$

From eq. (1), one may obtain the following equation, relating two stations along a Venturi tube,

$$\frac{u_1^2}{2g} + h_1 = \frac{u_2^2}{2g} + h_2 = \frac{u_n^2}{2g} + h_n . \quad (2)$$

Using Bernoulli's equation and the conservation of mass equation, one may be able to calculate velocities and pressures inside a Venturi tube with known geometry. Specifically, Bernoulli's equation may be used to obtain the difference in pressure heads between any point along the Venturi tube and the inlet point:

$$h_n - h_1 = \frac{u_1^2 - u_n^2}{2g} , \quad (3)$$

from which the dimensionless pressure difference can be derived, using the conservation of mass equation, as

$$\frac{h_n - h_1}{u_2^2/2g} = \frac{u_1^2 - u_n^2}{u_2^2} \approx \left(\frac{A_2}{A_1} \right)^2 - \left(\frac{A_2}{A_n} \right)^2 = \left(\frac{d_2}{d_1} \right)^4 - \left(\frac{d_2}{d_n} \right)^4 . \quad (4)$$

The right hand side of eq. (4) is a purely geometric expression and represents the **theoretical** or **ideal** dimensionless pressure difference. The left hand side of eq. (4) can be measured from the experiment and thus represents the **actual** dimensionless pressure difference.

Technical Details

The experiment's apparatus is shown in Figure 2, with the relevant measurements of the experimental apparatus shown in Figure 3.

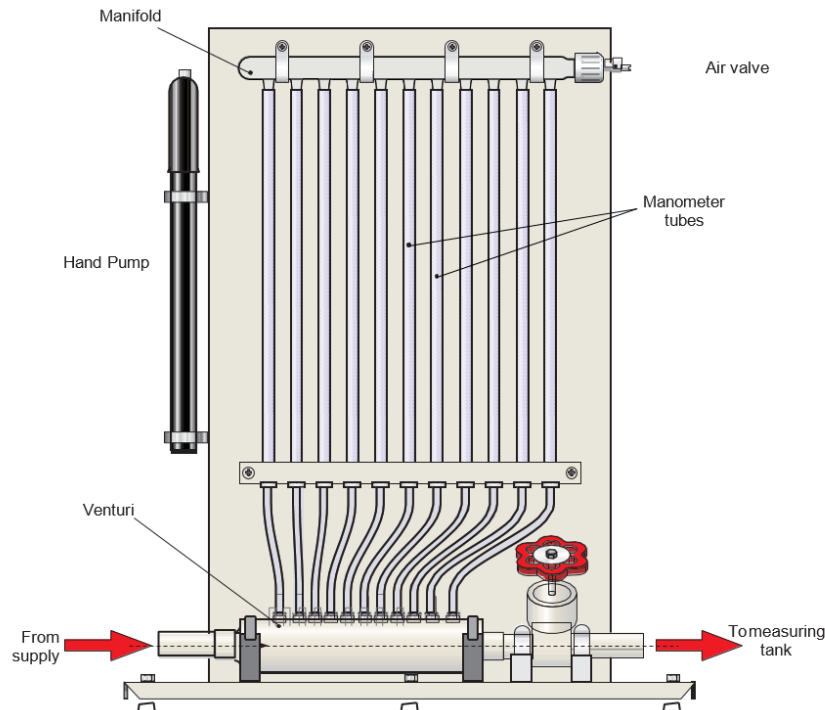
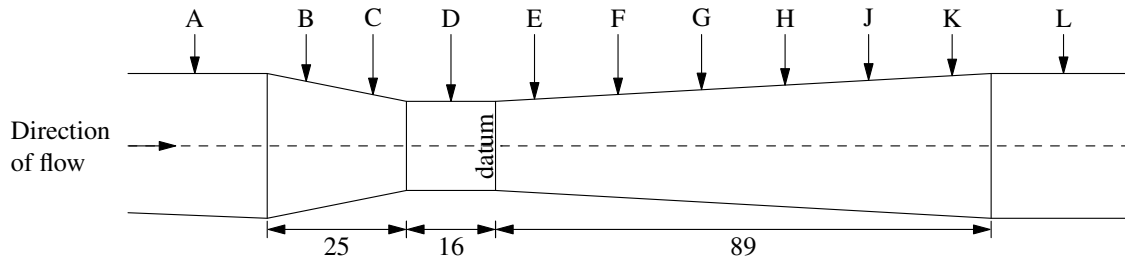


Figure 2: Apparatus

Experimental Procedure

The experiments were performed by the TAs ahead of the laboratory session.

1. Listening to your TAs instructions and make sure your apparatus is ready for the measurements by inspecting the following:
 - (a) No water at the top air chamber
 - (b) No water droplets throughout any of the vertical head measurement pipes (lightly tap the pipes until all droplets have disappeared, including from the base pipe sections)
 - (c) basin is drained and can be used to measure flow rate (alternatively - the basin is monitored for measuring a sufficiently large change in water level per unit time)
 - (d) all water head levels are within the manometer's readable range
 - (e) the outlet hose is level and not sunk in the water basin (extra 5 pts for an explanation as to why this note is important)
2. With the current flow rate setting, measure all [A]-[L] heads and fill the second column in Table 2.
3. Measure the flow rate for the current setting. Place the measured value at the top of Table 2.
4. Repeat previous steps with a significantly different (at least 0.1 L s^{-1}) flow rate value until you fill Table 3.



Piezometer Reference	A	B	C	D	E	F	G	H	J	K	L
Diameter (mm)	26.00	23.20	18.40	16.00	16.79	18.47	20.16	21.84	23.53	25.21	26.00
Area (mm ²)	530.9	422.7	265.9	201.1	221.4	267.9	319.2	374.6	434.8	499.2	530.9
Distance from datum (mm)	-54.0	-34.0	-22.0	-8.0	7.0	22.0	37.0	52.0	67.0	82.0	102.0

Figure 3: Cross section diameters and locations

Calculations

Note: You WILL be doing the following to complete this section of the laboratory:

- **Review the sample Excel file** provided to you on the course webpage or Blackboard, with entered data as a resource to guide your own data entries.
- **You will be entering the data gathered during the experiment into an empty Excel file** provided to you on the course webpage or Blackboard.
- **You will be running a Jupyter Notebook provided to you** to analyze and plot the data gathered from the experiment. **THIS step is NOT a group exercise.**
- **You must attach your Excel spreadsheet and a well-commented and clean Jupyter notebook for your final submission.**
- **You must ensure that all calculated data in all columns are displayed in the Jupyter notebook.**

Perform the following to complete this section of the laboratory:

1. Calculate the ideal dimensionless pressure for all measuring points using eq. (4) and fill in Table 1.
2. Calculate measured dimensionless pressure for all measuring points using eq. (4), and place the results into Table 2 and Table 3. Be sure to use the LHS of eq. (4), with measured values only.
3. On a single graph, plot the two actual and the theoretical dimensionless pressure values against distance from the Venturi tube inlet position (follow values for l in Figure 3). Ensure your figure is legible, easy to read, and interpretable. Also, note the scale, axes limits, line thicknesses, and overall figure presentation. All aspects will be graded.

Observations and Results

Table 1: Geometric measurements and ideal dimensionless (or dim-less) pressure

(Throat area)/(Upstream area) = $(d_2/d_1)^2$					
Tapping	l (mm)	d (mm)	A (mm ²)	d_2/d_n	Ideal dim-less pressure
A (upstream, h_1)					
B					
C					
D (throat, h_2)					
E					
F					
G					
H					
J					
K					
L					

Table 2: Calculations for flow rate-1

Flow, $Q_1 =$ [L s ⁻¹]				
Velocity, $u_2 =$ [m s ⁻¹]				
$u_2^2/2g =$				
Tapping	l (mm)	h_n (mm)	$h_n - h_1$ (mm)	Actual dim-less pressure
A (upstream, h_1)				
B				
C				
D (throat, h_2)				
E				
F				
G				
H				
J				
K				
L				

Table 3: Calculations for flow rate-2

Flow, $Q_2 =$ $[\text{L s}^{-1}]$				
Velocity, $u_2 =$ $[\text{m s}^{-1}]$				
$u_2^2/2g =$				
Tapping	l (mm)	h_n (mm)	$h_n - h_1$ (mm)	Actual dim-less pressure
A (upstream, h_1)				
B				
C				
D (throat, h_2)				
E				
F				
G				
H				
J				
K				
L				

Discussion

IMPORTANT: This portion of the laboratory work requires individual effort. Interaction with your group is NOT permitted. Please sign this academic integrity pledge below, acknowledging your individual effort. Without your signature, this portion will not be graded. The professor may use the “SafeAssign” feature on Blackboard to assess plagiarism.

According to the GWU Code of Academic Integrity, I pledge that I have neither given nor received unauthorized assistance on this work.

Last Name

First Name

Student's Signature

Date.

_____, _____, _____, _____

Answer each of the following questions

Write the five major assumptions associated with Bernoulli's equation. [5 pts]

Using Bernoulli's and the conservation of mass equations, derive an expression for u_2 .

Hint: Use the left-hand side of eq. (2) with the conservation of mass equation. [5 pts]

Briefly explain the points below based on the plot you made in the calculations section with reasons.

1. Variation of the manometer readings along the tube [5 pts]

2. Variation of the trend for the two different flow rates [5 pts]

3. Comment on any reasons that could cause your theoretical values and actual values to differ. [5 pts]

Deliverables Check-list

Read the submission instructions below carefully and complete them by the deadline mentioned on BlackBoard.

- ☐ Completed Tables 1, 2, and 3 and saved as a PDF. [10 pts each Table. Total=30 pts]
- ☐ Completed Excel File with the data gathered during the experiment and that can be processed with the Python program provided to you. [15 pts]
- ☐ Python Jupyter notebook (an *.ipynb file) adapted to the data you gathered and reported in Tables 1, 2, and 3. **Note:** Sample data and a Jupyter notebook were provided to you. [15 pts]
- ☐ Jupyter notebook will generate a plot of your data. Upload the figure in PNG-format [15 pts]
- ☐ Answer each of the following questions on a separate sheet of paper, not exceeding the space in this document.
 - ☐ Write the five major assumptions associated with Bernoulli's equation. [5 pts]
 - ☐ Using Bernoulli's and the conservation of mass equations, derive an expression for u_2 . Hint: Use the left-hand side of eq. (2) with the conservation of mass equation. [5 pts]
 - ☐ Briefly explain the points below based on the plot you made in the calculations section with reasons.
 - ☐ Variation of the manometer readings along the tube [5 pts]
 - ☐ Variation of the trend for the two different flow rates [5 pts]
 - ☐ Comment on any reasons that could cause your theoretical values and actual values to differ. [5 pts]

IMPORTANT: Your data entered in the Tables should be in PDF format. Data used in conjunction with the Python program should be an Excel file. Your answers to the discussion questions should be in PDF format. The Jupyter notebook should be an *.ipynb file. The plot should be saved in PNG format. Other formats will not be graded. Make one zip-file of the 4 submission files (2 PDFs, 1 Excel file, 1 *.ipynb file, and 1 PNG file) in the specified format only. **Upload to Blackboard the final deliverables as one zip-file. Hit 'Submit' button on Blackboard to complete your submission. The zip-file will not register as submitted, unless you do so.**